

March 2021

MANUFACTURING PROCESSES – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty): 1x20
- i) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - ii) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - iii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - iv) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - v) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - vi) Above _____ temperature, working of metals is named hot working.
 - vii) Drawing is done to _____ the diameter of wire.
 - viii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - ix) Accurate taper and step tapers can be machined with _____ attachment.
 - x) Finishing and sizing a hole is called _____.
 - xi) Spot welding is one of the _____ welding processes.
 - xii) The revolving cutter of a milling machine is held on a _____.
 - xiii) In universal milling machine table can have the _____ arrangement about a vertical axis.
 - xiv) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - xv) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - xvi) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - xvii) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - xviii) A twist drill has _____ flutes.
 - xix) Pattern used in precision investment casting normally made of _____.

- xx) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xxi) MIG welding uses _____ electrode.
- xxii) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.

2. What is core print? What is core? Write the advantages of wood as pattern making material. 2+3+5
 3. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 4. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 5. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 6. Write the functions of different parts of a radial drilling machine. 10
 7. Describe hot chamber and cold chamber die-casting method. 5+5
 8. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
 9. Describe submerged arc welding process. 10
 10. Describe different types of milling cutter. 10
 11. Write short notes on different types of chips. 10
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